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Predicting breast cancer response to neoadjuvant therapy by integrating radiomic and deep-learning features from early-and-peak phases of DCE-MRI

Yabin Zhang^{1,2†}, Jiumei Cai^{3†}, Chunxiao Cui⁴, Shouliang Qi^{1,2*} and Dan Zhao^{3*}

Abstract

Background Non-invasive prediction of pathological complete response (pCR) in breast cancer patients undergoing neoadjuvant therapy (NAT) is crucial for adjusting surgical strategies and optimizing treatment plans. This study aims to develop a predictive model that integrates traditional radiomics and 3D deep learning features from early and peak phases of dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) to predict pCR after NAT.

Methods This retrospective study included 234 breast cancer patients from two hospitals who received neoadjuvant therapy. Dataset 1 ($n=204$) was used for model development, and Dataset 2 ($n=30$) was used for external validation. Traditional radiomics and 3D deep learning features were extracted from both the whole DCE-MRI image and the tumor region of interest (ROI). Features from different sources were integrated, followed by feature selection using independent sample t-tests and least absolute shrinkage and selection operator (LASSO) regression, and the top ten discriminative features were selected for model training. Logistic regression was used to build predictive models, and their performance was evaluated using receiver operating characteristic curves and area under curve (AUC). The DeLong test was used to assess differences between the AUC values of different models, and SHAP (SHapley Additive exPlanations) analysis was employed to examine the relationship between the features of the models and pCR.

Results For models using only traditional radiomics features, the combined model integrating early and peak phases of DCE-MRI provided the best pCR prediction. The performance of this combined model was further enhanced by adding 3D deep learning features. The optimal model (RD_EP), which integrated radiomics and deep learning features from early and peak phases of DCE-MRI, achieved AUC values of 0.892 (95% CI: 0.853–0.922) on Dataset 1 and 0.825 (95% CI: 0.713–0.886) on Dataset 2. The DeLong test showed that RD_EP had statistically significant differences compared to other prediction models ($p < 0.05$). SHAP analysis demonstrated that two radiomics texture features contributed the most to the model.

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Conclusion Integrating traditional radiomics and 3D deep learning features from different phases of DCE-MRI can accurately predict pCR to NAT in breast cancer accurately. Multi-phase imaging and diverse features are important to improve predictive accuracy and the constructed model may help guide personalized treatment strategies.

Keywords Breast cancer, Magnetic resonance imaging, Radiomics, Deep learning, Feature integration, Neoadjuvant therapy, Pathological complete response

Introduction

Breast cancer is currently one of the most common cancers worldwide and a major cause of cancer-related mortality among women [1, 2]. Neoadjuvant therapy (NAT) is an essential treatment modality for early high-risk and locally advanced breast cancer. It can reduce tumor size, lower Tumor-Node-Metastasis (TNM) staging, improve surgical success rates, increase breast-conserving surgery opportunities, reduce distant metastases and guide subsequent treatments [3–6]. Some patients achieve pathological complete response (pCR) through NAT, and compared to non-pCR patients, they have significantly better long-term prognoses [7]. Therefore, predicting pCR in breast cancer patients is crucial for formulating precise treatment plans and improving patient outcomes [8].

In clinical practice, imaging examinations are the most common methods for preoperatively evaluating the efficacy of NAT in breast cancer patients, including mammography, ultrasound, magnetic resonance imaging (MRI), diffuse optical spectroscopy, and Positron Emission Tomography – Computer Tomography (PET-CT) [9–12]. Among these, breast MRI is currently the most accurate non-invasive method for predicting pCR, closely aligning with pathological evaluations [10, 11, 13, 14]. Particularly, dynamic contrast-enhanced MRI (DCE-MRI) can clearly depict tumor boundaries, morphology, structure, and internal hemodynamics [15, 16]. These features are crucial for the quantitative assessment of tumors. With the advancement of radiomics, high-dimensional quantitative analysis of medical imaging data allows for the extraction of numerous quantitative features, which can be used in disease diagnosis, prognosis prediction, and treatment response evaluation [17–20], holding significant clinical importance.

Radiomics can effectively predict the efficacy of NAT in breast cancer patients [21–25]. Eun et al. [22] used commercial software to perform texture analysis on patients' MRI images and established a pCR prediction model based on texture features. Zheng et al. [24] used features from three different regions of breast tissue to predict pCR in breast cancer patients, highlighting the potential influence of the tumor microenvironment on NAT efficacy. Sutton et al. [26] employed automated radiomics methods, including semi-automatic tumor segmentation, to develop a recursive feature elimination-random forest classifier for predicting pCR. However, in their studies, most features were derived from specialized

feature extraction software, including Pyradiomics [27], TexRAD [28], IBEX [29], MaZda [30], and LIFEx [31], with these features primarily being manually predefined. Deep learning algorithms, on the other hand, can automatically extract complex high-dimensional features from medical images, capturing tumor information that traditional radiomics features might overlook, and have already been applied in breast cancer detection, diagnosis, and prediction [32, 33]. Li et al. [34] demonstrated the effectiveness of a fusion model based on traditional radiomics and deep learning features in predicting axillary lymph node metastasis in breast cancer. Peng et al. [35] applied deep learning methods to predict pathological response to NAT in breast cancer patients and found that these methods outperformed traditional radiomics approaches. However, the deep learning features in their study were extracted only from 2D slices of the tumor, inevitably losing some important 3D information. Three-dimensional feature extraction can capture more spatial information of the tumor, providing a more comprehensive analysis. Integrating 3D deep learning features with traditional radiomics features for pCR prediction in breast cancer patients is a promising and challenging research direction.

Radiomics models based on DCE-MRI can evaluate the efficacy of neoadjuvant therapy in breast cancer patients [21–25, 36], predict HER2 and Ki-67 levels [37, 38], sentinel lymph node metastasis status [39, 40], and distinguish tumor molecular subtypes [41, 42]. However, these studies extracted tumor features only from the early [22] or peak [23] phases of dynamic contrast-enhanced imaging. Huang et al. [42] used different phases of DCE-MRI to differentiate breast cancer molecular subtypes, achieving promising results. This indicates the potential clinical significance and efficacy of DCE-MRI at different phases. Currently, there is no consensus on which phase of DCE-MRI provides the best predictive performance for pCR.

This study proposes to use baseline DCE-MRI early and peak phases images, extract and integrate radiomics and 3D deep learning features, and establish machine learning models of predicting pCR to NAT in breast cancer. By comparing the performance of different prediction models, the study aims to identify the optimal feature combinations for predicting NAT efficacy in patients. This will help clinical doctors in providing more accurate personalized treatment plans for breast cancer patients. The main innovations and contributions of this

study are: (a) An integrative model is constructed, which can accurately predict neoadjuvant therapy response in breast cancer by DCE-MRI. (b) From a methodological viewpoint, radiomics and 3D deep learning features from the early and peak phases of DCE-MRI are analyzed, clarified, and integrated. (c) A group of imaging biomarkers, which are predictive for neoadjuvant chemotherapy response in breast cancer, is created. (d) The evaluation on an external dataset has proved the great generalization capacity of the developed predictive model.

Materials and methods

Patients and data

This study is a retrospective study based on data from two hospitals (Hospital A: Liaoning Cancer Hospital, Hospital B: Affiliated Hospital of Qingdao University). The study has been approved by the medical ethics committees of both hospitals (No. 20221101; QYFY WZLL 28631), with waived patient informed consent. We retrospectively collected clinical data of female breast cancer patients diagnosed pathologically and undergoing neoadjuvant therapy before surgery, ranging from stage I to III, from Hospital A (January 2018 to December 2022) and Hospital B (July 2021 to February 2023).

The inclusion and exclusion criteria are given as follows. Inclusion criteria: (a) Initially diagnosed with primary unilateral breast cancer, with a single lesion or accompanied by satellite lesions (based on MRI diagnosis before neoadjuvant therapy), without distant metastasis, and no history of radiation therapy, surgery, or other breast cancer treatments; (b) Received at least 4 cycles of NAT; (c) Underwent surgical treatment after completion of NAT; (d) Complete pathological and pre-NAT biopsy data available post-surgery; (e) Pathological type is invasive ductal carcinoma. Exclusion criteria: (a) Bilateral breast cancer; (b) History of distant metastasis or other malignant tumors; (c) Incomplete pathological, clinical, or imaging data.

According to the inclusion and exclusion criteria, Hospital A collected 204 cases of enrolled patients, and Hospital B collected 30 cases. The cases from Hospital A (Dataset 1) were randomly split into training and testing sets at a ratio of 7:3, the cases from Hospital B (Dataset 2) were used as an external validation set. The flow of patient enrollment is shown in Fig. 1.

Figure 2 shows the overall workflow of this study. Firstly, DCE-MRI data were acquired and images were pre-processed. Then, traditional radiomics and 3D deep learning features were extracted from early and peak phases DCE-MRI images. Next, the feature selection was performed using independent sample t-tests and least absolute shrinkage and selection operator (LASSO) regression. Finally, prediction models were established and a validation assessment was conducted.

MRI acquisition and tumor segmentation

All enrolled cases underwent contrast-enhanced breast MRI examination before neoadjuvant therapy. For DCE-MRI scanning, Hospital A used a 1.5T GE Signa MRI scanner, while Hospital B used 3.0T SIEMENS Skyra and 3.0T GE Discovery MRI scanners. After the mask scanning, contrast-enhanced scanning was performed by intravenous bolus injection of contrast agent at a dose of 0.1mmol/kg using a high-pressure injector. Hospital A used Omniscan (gadodiamide, GE Healthcare, Ireland) as the contrast agent, whereas Hospital B used Magnevist (gadopentetate dimeglumine, Bayer AG, Germany). Post-enhancement, the 1.5T GE Signa MRI of Hospital A conducted 8 consecutive non-interval scans with a slice thickness of 3.5 mm, slice spacing of 0 mm, and each scan lasting 58 s. The 3.0T SIEMENS Skyra MRI of Hospital B performed 5 consecutive non-interval scans with a slice thickness of 1 mm, slice spacing of 0.2 mm, and each scan lasting 90 s. The 3.0T GE Discovery MRI of Hospital B executed 5 consecutive non-interval scans with a slice thickness of 1 mm, slice spacing of 0.5 mm, and each scan lasting 65 s.

Pre-neoadjuvant therapy DCE-MRI images of enrolled cases were downloaded from the Picture Archiving and Communication Systems, exported, and saved in DICOM format for medical digital imaging. Based on the time-signal intensity curve of each case's DCE-MRI, early and peak phases sequence images were exported. Two radiologists with 3 years and more than 5 years of breast imaging diagnosis experience used ITK-SNAP (www.itk-snap.org, Version 4.0.1) image processing software to manually delineate the regions of interest (ROIs) around lesions on each transverse axial slice of breast cancer tumors, and then automatically generated the entire lesion's ROI. Subsequently, a radiologist with over 15 years of breast imaging diagnosis experience verified the regions of interest and their locations. To further assess inter-reader reproducibility, we calculated the Dice similarity coefficient between the two radiologists' annotations, yielding a mean value of 0.836 (95% CI: 0.802–0.867), indicating good consistency of ROI delineation.

Image pre-processing

The image preprocessing in this study included N4 bias field correction [43] and cropping based on lesion regions. The preprocessing results are shown in Fig. 3.

Bias field effect is a low-frequency spatial artifact commonly observed in MRI, causing non-uniform intensity values in images, thereby affecting subsequent image analysis and processing. In this study, N4 bias field correction was applied to the patients' MRI images. N4ITK Bias Field Correction is an algorithm used in medical image processing to correct bias field effects, aiming to

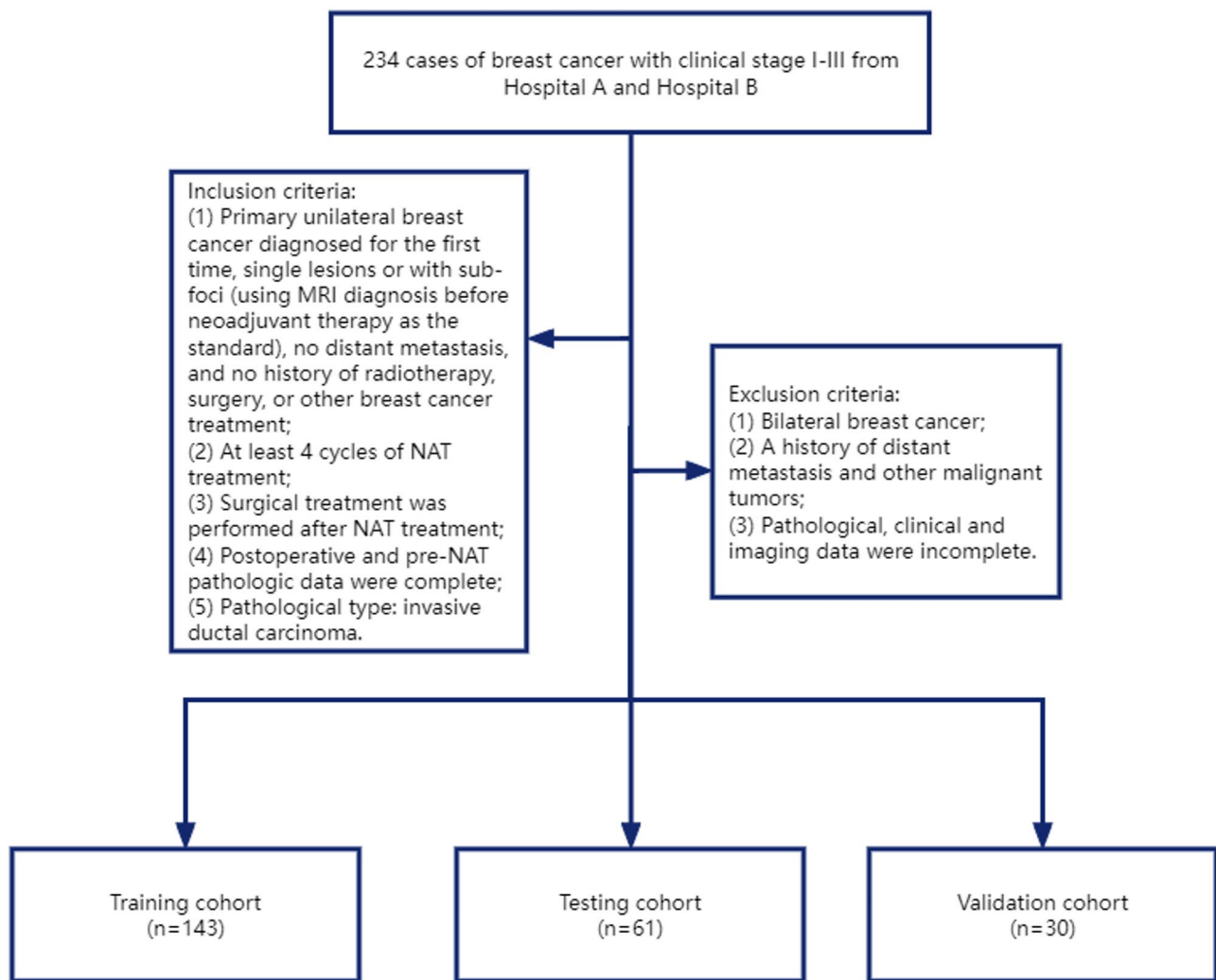


Fig. 1 Flowchart of case inclusion, exclusion, and data grouping

improve image quality and the accuracy of subsequent analysis.

The original MRI images of patients are large in size ($512 \times 512 \times 48$), with the lesion regions occupying only a small portion. For ease of feature extraction, we performed cropping of MRI images based on the lesion regions. We used the Python-based MONAI toolkit (<https://monai.io/>, Version 1.2.0) to crop the images according to the regions of interest (ROI) delineated by the physicians. The mask files were also cropped to the corresponding size for subsequent feature extraction.

Additionally, to reduce the impact of inter-centre differences in magnetic field strength and contrast protocols on feature consistency and model performance, we applied uniform resampling and z-score normalization before feature extraction and executed the same pipeline across centres, thereby reducing heterogeneity and preserving the independence of the external validation dataset. Specifically, after N4 bias-field correction,

all volumes were resampled to a fixed voxel grid (harmonized voxel spacing and matrix size); image volumes used trilinear interpolation, and masks used nearest-neighbour interpolation to preserve discrete labels. We then performed z-score normalization on each volume individually, computing the mean and standard deviation from that volume's effective voxels (non-zero voxels) and scaling intensities accordingly. For multi-phase DCE-MRI, normalization was performed separately for each phase to avoid smoothing or equalizing dynamic signals across phases.

Feature extraction

Based on the cropped DCE-MRI images, we performed feature extraction on both the entire 3D image (image-based) and the region of interest (ROI-based) 3D images.

Extraction of traditional radiomics features: This study utilized Pyradiomics (<https://pyradiomics.readthedocs.io/en/latest/index.html>, version 3.0.1) to extract traditional

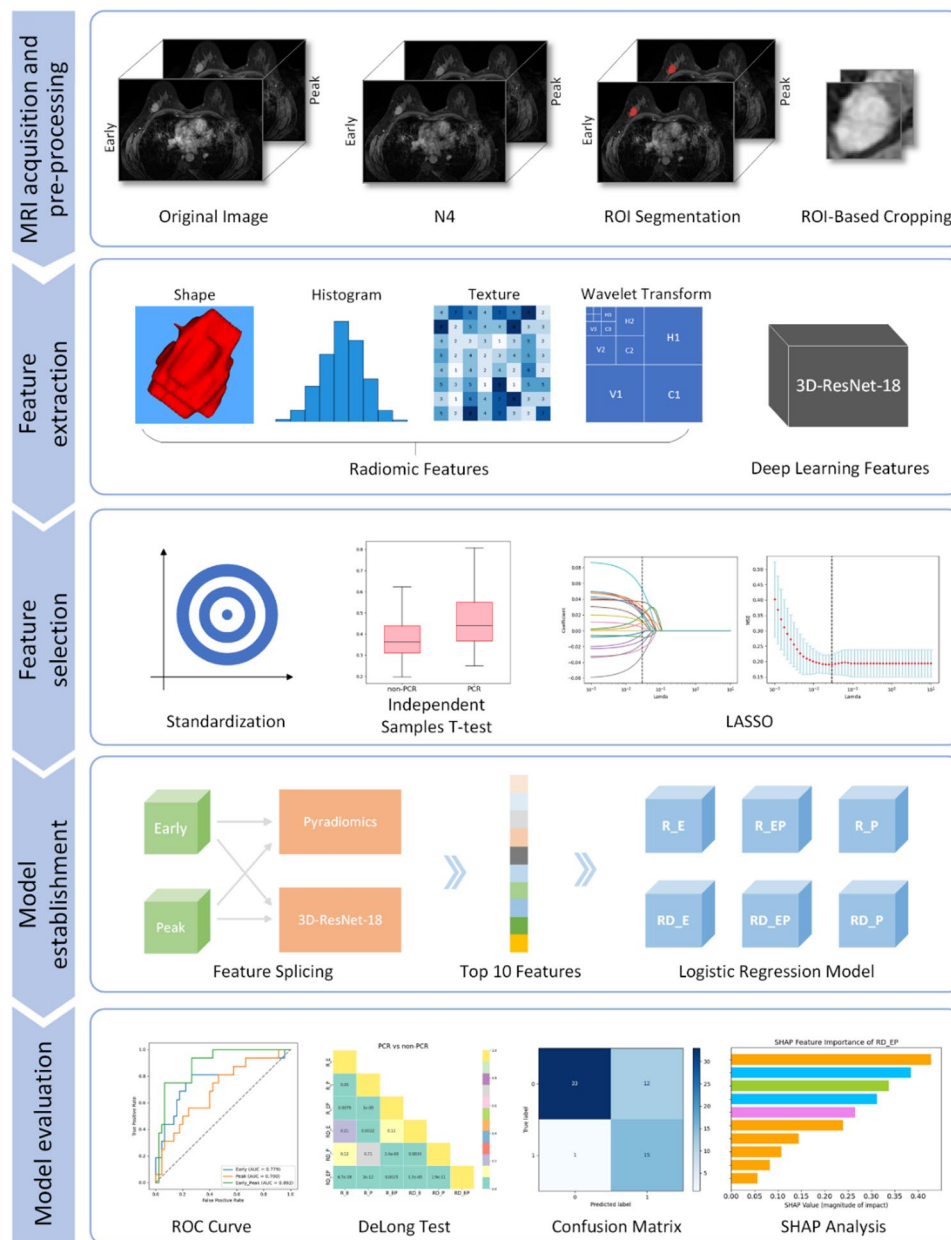


Fig. 2 Experimental workflow, including data acquisition and preprocessing, feature extraction, feature selection, model establishment, and evaluation

radiomics features from the images. We extracted 1409 traditional radiomics features separately from the early and peak phases of DCE-MRI, including First Order Features, Shape Features, Gray Level Co-occurrence Matrix (GLCM), Gray Level Size Zone Matrix (GLSZM), Gray Level Run Length Matrix (GLRLM), Neighboring Gray Tone Difference Matrix (NGTDM), and Gray Level Dependence Matrix (GLDM). For detailed definitions and explanations of these features, please refer to the Pyradiomics official documentation (<https://pyradiomics.readthedocs.io/en/latest/features.html>), which provides comprehensive information and detailed explanations.

Extraction of 3D deep learning features: We utilized the pre-trained 3D-ResNet-18 deep learning network (<https://github.com/pytorch/vision/blob/main/torchvision/models/video/resnet.py>) to extract 512 3D deep learning features from both early and peak phases of DCE-MRI.

The 3D-ResNet-18 is a deep convolutional neural network architecture based on the ResNet (Residual Network) framework, specifically designed for feature extraction and classification tasks with three-dimensional (3D) volume data. Unlike traditional ResNet architectures, 3D-ResNet-18 incorporates 3D convolution operations, which enhance its capability to effectively process 3D volume data such as CT or MRI images [44]. These 3D

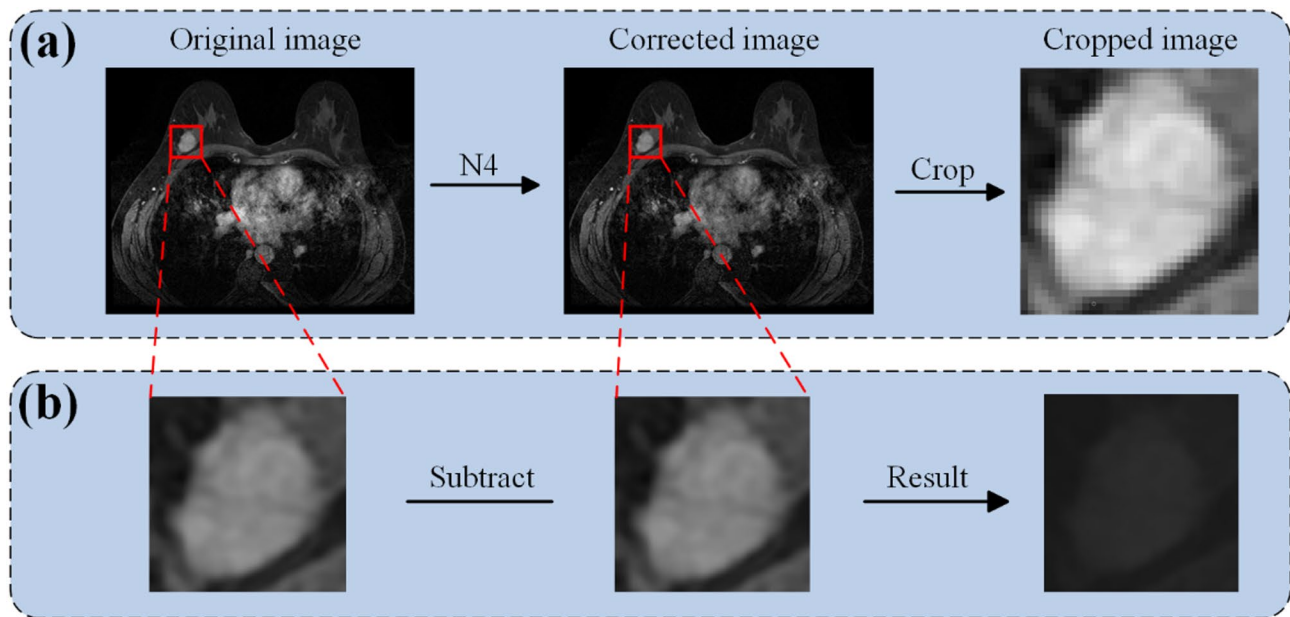


Fig. 3 Image preprocessing includes N4 bias field correction and image cropping. (a) Image preprocessing; (b) Results of N4 bias field correction on lesion regions

convolution operations aid in capturing and understanding spatial information within images, which is crucial in medical image processing. They enable more accurate extraction of complex 3D structures and detailed features, thereby supporting precise diagnosis, analysis, and prediction.

To adapt the pretrained 3D-ResNet-18 model for medical imaging, we modified its first convolutional layer to accept a single-channel input (instead of the original three RGB channels) and initialized the weights by averaging the pretrained kernels across channels. The final fully connected classification layer was removed, retaining only the convolutional backbone for feature extraction. All DCE-MRI images were resampled to a fixed size and standardized using z-score normalization. For ROI-based extraction, tumor segmentation masks were applied to ensure that features were derived exclusively from the lesion region. The processed images were then passed through the modified 3D-ResNet-18, yielding a 512-dimensional feature vector for each case that captures high-level semantic, spatial, and morphological characteristics of the tumor.

Feature selection

The extracted traditional radiomics and 3D deep learning features in this study were numerous. Using all features directly to build prediction models may lead to the “curse of dimensionality” [45], resulting in excessive model complexity and a risk of overfitting. Therefore, identifying the most valuable features is crucial.

This procedure was conducted solely on the training set, excluding the test set. Firstly, independent sample t-tests were used to verify if each feature exhibited significant differences between the pCR and non-pCR sample sets. Features with a p-value less than 0.05 were deemed to have significant differences and were included in the selected feature list. Subsequently, the LASSO with ten-fold cross-validation was employed to analyze the feature list obtained from independent sample t-tests, selecting the optimal subset of features. LASSO extends linear regression by adding L1 regularization, which automatically shrinks the regression coefficients of certain features to zero during the fitting process. This process removes unimportant and redundant features, hereby achieving feature selection.

In total, 1,409 traditional radiomics features and 512 three-dimensional deep learning features were extracted separately from the early and peak phases of DCE-MRI. Six feature combinations were defined:

- a. (a) R_E: Radiomics_Early phase features;
- b. (b) R_P: Radiomics_Peak phase features;
- c. (c) R_EP: Radiomics_Early and Peak phases features;
- d. (d) RD_E: Radiomics and Deep Learning_Early phase features;
- e. (e) RD_P: Radiomics and Deep Learning_Peak phase features;
- f. (f) RD_EP: Radiomics and Deep Learning_Early and Peak phases features.

During the LASSO procedure, coefficients were computed and features ranked by absolute magnitude to evaluate their importance. For each phase (Early or Peak) and feature type (Radiomics or Deep Learning), the top ten features were selected for subsequent model training.

Model establishment and evaluation

This study used the Logistic Regression (LR) machine learning model as the classifier, provided by the open-source Python library Scikit-Learn (Version 1.2.2). Based on different feature combinations, six binary classification models were developed: Radiomics_Early phase model (R_E), Radiomics_Peak phase model (R_P), Radiomics_Early and Peak phases model (R_EP), Radiomics and Deep Learning_Early phase model (RD_E), Radiomics and Deep Learning_Peak phase model (RD_P), and Radiomics and Deep Learning_Early and Peak phases model (RD_EP). Among these, R_E, R_P,

and R_EP are models using only radiomics features, while RD_E, RD_P, and RD_EP are models using a combination of radiomics and deep learning features. The model development process is illustrated in Fig. 4.

We used the Synthetic Minority Over-sampling Technique (SMOTE) method to address the issue of class imbalance in our data. SMOTE balances imbalanced datasets by generating synthetic samples through interpolation among minority class samples [46]. Its advantage lies in its ability to significantly increase the number of minority class samples, balance the class distribution and improve the classifier's ability to recognize minority classes, thus improving the overall performance of the model. For the selection of hyperparameters in the logistic regression model, we used grid search with 10-fold cross-validation to determine the optimal parameters for the model.

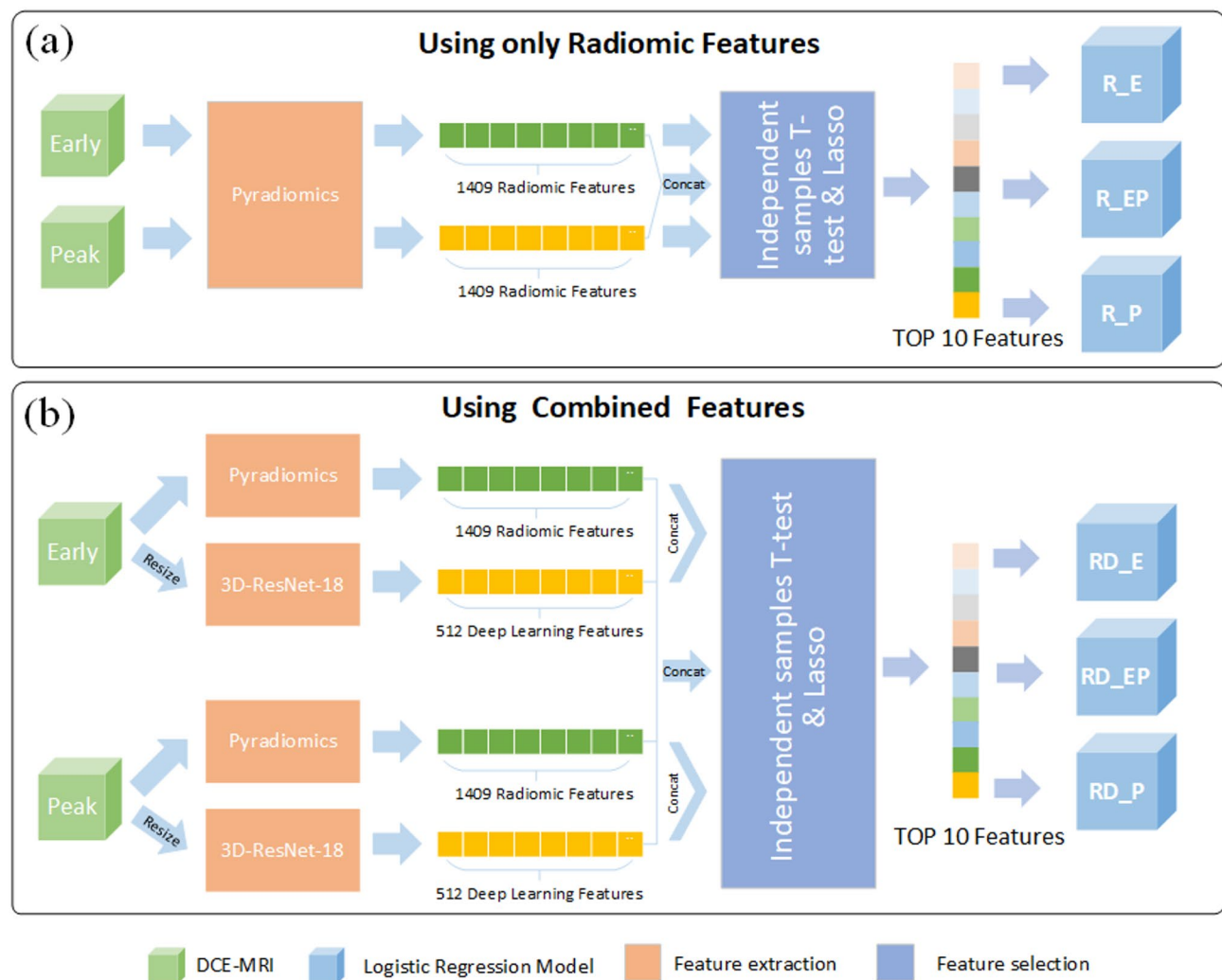


Fig. 4 The process of model establishment. **a** Models using only traditional radiomics features; **(b)** Combined models using traditional radiomics and 3D deep learning features

In this study, Dataset 1 was used for prediction model development and evaluation, Dataset 2 was used for external validation to assess the generalization ability of the models. We used metrics such as specificity, sensitivity, accuracy, and area under the curve (AUC) to assess the performance of the model. Additionally, we plotted receiver operating characteristic (ROC) curves and confusion matrices. The DeLong test is used to assess whether there is a significant difference in performance between two models. A smaller p-value indicates a more significant difference between the two AUCs. When the p-value is less than 0.05, it indicates a significant difference between the two models. SHapley Additive exPlanations (SHAP) is used to analyze the relationship between different features in the model and the prediction results, determining the contribution of each feature to the model [47].

Statistical analysis

All experiments in this study, including statistical analysis, feature extraction and selection, model establishment and model evaluation, were conducted using Python (Version 3.11.4), implemented on PyCharm (Version 2023.2.1). For continuous variables, we performed normality testing using the Kolmogorov-Smirnov test. Then, depending on the results of the normality test, we conducted statistical analysis using independent sample t-tests or Mann-Whitney U tests. For categorical variables, we employed chi-square tests and Fisher's exact tests for statistical analysis. We employed the DeLong test to analyze differences in AUC values among different classification models. If the p-value is less than 0.05, it indicates that there is a significant difference in the AUC between different models, demonstrating statistical

significance. The interpretability analysis of the model was performed using the Python toolkit SHAP (<https://github.com/slundberg/shap>). 95% confidence intervals (CIs) for AUC, sensitivity, and specificity were obtained via a non-parametric, patient-level bootstrap (B = 2,000; percentile 2.5–97.5%). Sensitivity and specificity were computed at the pre-specified operating threshold (0.50).

Results

Patient characteristics

Table 1 shows the demographic and clinical characteristics analysis results of patients in datasets 1 and 2. In Dataset 1 ($n=204$), there were no significant differences between the pCR group ($n=53$) and non-pCR group ($n=151$) in terms of age, lesion side, T stage, N stage, ER (Estrogen receptor) status, PR (Progesterone receptor) status, and HER-2 (Human epidermal growth factor receptor 2) status. However, there was a significant difference in Ki-67 status between the two groups ($p=0.028$). In Dataset 2 ($n=30$), all demographic and clinical characteristics showed no significant differences between the pCR group and non-pCR group.

Features characteristics

After integrating traditional radiomics and deep learning features from early and peak phases of DCE-MRI, there are a total of 3842 features. After applying independent sample t-tests for dimensionality reduction, 489 features showed significant differences. Among these, 236 were from the early phase (212 radiomics features, 14 deep learning features), and 263 were from the peak phase (233 radiomics features, 30 deep learning features). The process of further feature selection using LASSO ultimately resulted in the selection of 65 features. Among these, 29

Table 1 Clinical characteristics of datasets 1 and 2

Characteristics		Dataset 1 ($n=204$)				Dataset 2 ($n=30$)			
		ALL ($n=204$)	pCR ($n=53$)	non-pCR ($n=151$)	p value	ALL ($n=30$)	pCR ($n=9$)	non-pCR ($n=21$)	p value
Age		52.520 ± 9.517	52.302 ± 10.013	52.596 ± 9.370	0.852	49.967 ± 8.202	47.778 ± 4.816	50.905 ± 9.230	0.235
Lesion side	L	99 (48.53%)	22 (41.51%)	77 (50.99%)	0.055	20 (66.67%)	7 (77.78%)	13 (61.90%)	1
	R	105 (51.47%)	31 (58.49%)	74 (49.01%)		10 (33.33%)	2 (22.22%)	8 (38.10%)	
T stage	T1-T2	174 (85.29%)	49 (92.45%)	125 (82.78%)	0.272	25 (83.33%)	8 (88.89%)	17 (80.95%)	1
	T3-T4	30 (14.71%)	4 (7.55%)	26 (17.22%)		5 (16.67%)	1 (11.11%)	4 (19.05%)	
N stage	N0-N1	171 (83.82%)	46 (86.79%)	125 (82.78%)	0.588	27 (90.00%)	8 (88.89%)	19 (90.48%)	1
	N2-N3	33 (16.18%)	7 (13.21%)	26 (17.22%)		3 (10.00%)	1 (11.11%)	2 (9.52%)	
ER status	Negative	80 (39.22%)	30 (56.60%)	50 (33.11%)	0.151	10 (33.33%)	4 (44.44%)	6 (28.57%)	0.167
	Positive	124 (60.78%)	23 (43.40%)	101 (66.89%)		20 (66.67%)	5 (55.56%)	15 (71.43%)	
PR status	Negative	99 (48.53%)	36 (67.92%)	63 (41.72%)	0.557	11 (36.67%)	5 (55.56%)	6 (28.57%)	1
	Positive	105 (51.47%)	17 (32.08%)	88 (58.28%)		19 (63.33%)	4 (44.44%)	15 (71.43%)	
HER-2 status	Negative	133 (65.20%)	28 (52.83%)	105 (69.54%)	0.339	14 (46.67%)	0 (0.00%)	14 (66.67%)	1
	Positive	71 (34.80%)	25 (47.17%)	46 (30.46%)		16 (53.33%)	9 (100.00%)	7 (33.33%)	
Ki-67 status	Negative	51 (25.00%)	7 (13.21%)	44 (29.14%)	0.028	9 (30.00%)	4 (44.44%)	5 (23.81%)	1
	Positive	153 (75.00%)	46 (86.79%)	107 (70.86%)		21 (70.00%)	5 (55.56%)	16 (76.19%)	

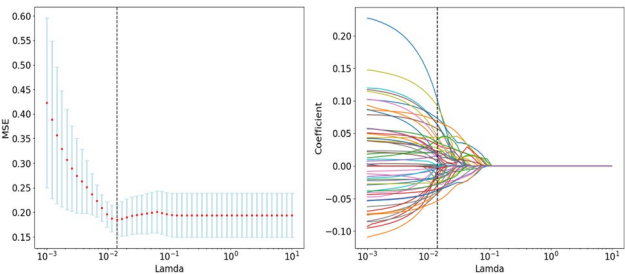


Fig. 5 The MSE and coefficients of each feature in LASSO as they vary with Lambda

features were from the early phase (23 radiomics features, 6 deep learning features), and 36 features were from the peak phase (28 radiomics features, 8 deep learning features). Figure 5 shows the mean squared error (MSE) of Lambda in the LASSO process and the variation of each feature coefficient with Lambda.

Based on the absolute values of the selected feature coefficients, the top 10 features were chosen for subsequent model training. These included 7 features from the early phase (6 radiomics features and 1 deep learning feature) and 3 features from the peak phase (2 radiomics features and 1 deep learning feature). Figure 6 shows the differences (mean and standard deviation) and the distribution of these 10 features between the pCR and non-pCR groups. In Fig. 6, the first two letters of each feature name represent the source of the feature: E stands for early phase, P stands for peak phase, R denotes radiomics features, and D denotes deep learning features. Except for PR_wavelet-LHH_glrmlm_HighGrayLevelRunEmphasis and ED_Feature246, the mean values of the rest of the features in the pCR group were higher than those in the non-pCR group.

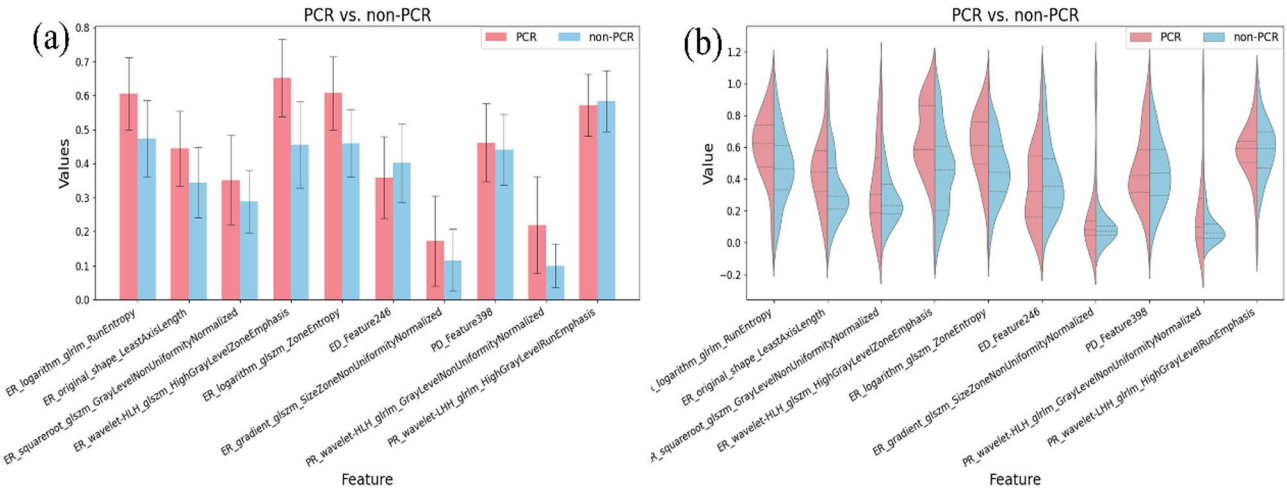


Fig. 6 Comparison of the top 10 features selected by LASSO. **a** Mean and standard deviation; **(b)** Data distribution

Performance of models using only radiomics features

Figure 7 (a) and (b) respectively show the ROC curves for models using only traditional radiomics features, based on image-based features and ROI-based features. It can be seen that the AUC values of the early phase, peak phase and early combined with peak phase combination models based on ROI-based features are higher than those based on image-based features. Specifically, the AUC range for the models based on image-based features is 0.685 to 0.743, while the AUC range for the models based on ROI-based features is 0.689 to 0.803.

Table 2 shows the performance metrics of each model. For the image-based models, the early and peak phases (R_EP) results are superior to those of the early phase (R_E) and peak phase (R_P) individually. Its sensitivity is 0.625 (95% CI: 0.455–0.761), specificity is 0.667 (95% CI: 0.614–0.783), accuracy is 0.656, and AUC is 0.743 (95% CI: 0.660–0.780). Similarly, for the ROI-based models, the R_EP results are the best. Its sensitivity is 0.875 (95% CI: 0.735–0.941), specificity is 0.711 (95% CI: 0.686–0.750), accuracy is 0.754, and AUC is 0.803 (95% CI: 0.712–0.851). Table 2 also shows that for both image-based and ROI-based models, the results of R_E are better than those of R_P. Bold values in the table indicate the best-performing results among all compared models.

In summary, when using only traditional radiomics features, the ROI-based models consistently outperform the image-based models. The early and peak phases (R_EP) model achieves the best results, with an AUC value of 0.803 (95% CI: 0.712–0.851).

Performance of models using combined radiomic and deep learning features

Based on models using only traditional radiomics features, we incorporated deep learning features to develop

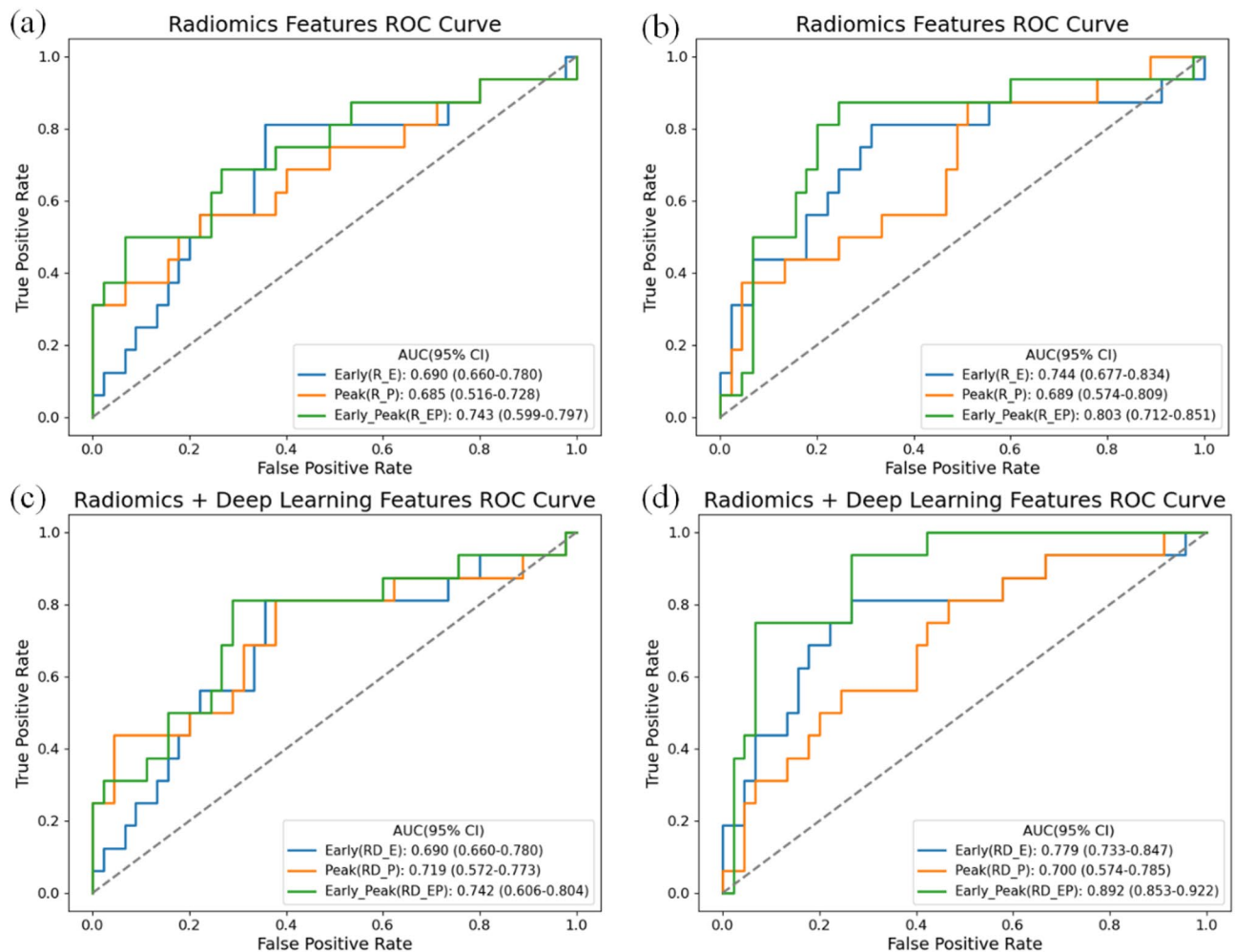


Fig. 7 ROC curves for all models. **a** Image-based radiomics feature-only models; **b** ROI-based radiomics feature-only models; **c** Image-based radiomics and deep learning feature combined models; **d** ROI-based radiomics and deep learning feature combined models

combined models using both radiomics and deep learning features. Similar to the models only using radiomics features, we also developed models using both image-based features and ROI-based features.

Figures 7 (c) and (d) display the ROC curves for the combined models using radiomics and deep learning features. Similar to the models only using radiomics features, the AUC values for the early phase, peak phase, and early combined with peak phase combination models based on ROI-based features are generally higher than those based on image-based features. Specifically, the AUC range for the image-based models is 0.690 to 0.742, while the AUC range for the ROI-based models is 0.700 to 0.892.

Table 3 shows the results for the combined models using radiomics and deep learning features. For the image-based feature models, the early and peak phases (RD_EP) model outperforms the individual early (RD_E) and peak phase (RD_P) models. Its Sensitivity is 0.813 (95% CI: 0.653–0.899), Specificity is 0.622 (95% CI:

0.529–0.692), Accuracy is 0.672, and AUC is 0.742 (95% CI: 0.606–0.804). Similarly, for the ROI-based feature models, the RD_EP model also delivers the best results, with a Sensitivity of 0.938 (95% CI: 0.886–0.952), Specificity of 0.733 (95% CI: 0.590–0.859), Accuracy of 0.787, and AUC of 0.892 (95% CI: 0.853–0.922). It is noteworthy that, for the image-based feature models, RD_P performs better than RD_E, whereas for the ROI-based feature models, RD_E performs better than RD_P. Bold values in the table indicate the best-performing results among all compared models.

From the perspective of feature integration, adding deep learning features into the combined models results in improved performance compared to models using only traditional radiomics features. Among these, the best-performing model is the ROI-based radiomics and deep learning features from early and peak phases model (RD_EP), which achieves an AUC of 0.892(95% CI: 0.853–0.922) and an accuracy of 0.787.

Table 2 Results of models using only traditional radiomics features

Feature source	Model	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy	AUC (95% CI)
Image-Based	RD_E	0.625 (0.455–0.761)	0.667 (0.614–0.783)	0.656	0.690 (0.660–0.780)
	RD_P	0.563 (0.433–0.677)	0.622 (0.573–0.748)	0.607	0.685 (0.516–0.728)
	RD_EP	0.688 (0.476–0.899)	0.644 (0.592–0.693)	0.656	0.743 (0.599–0.797)
ROI-Based	RD_E	0.750 (0.637–0.895)	0.689 (0.584–0.781)	0.705	0.744 (0.677–0.834)
	RD_P	0.563 (0.483–0.743)	0.600 (0.526–0.633)	0.590	0.689 (0.574–0.809)
	RD_EP	0.875 (0.735–0.941)	0.711 (0.686–0.750)	0.754	0.803 (0.712–0.851)

Table 3 Results of the combined models

Feature source	Model	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy	AUC (95% CI)
Image-Based	RD_E	0.625 (0.455–0.761)	0.667 (0.614–0.783)	0.656	0.690 (0.660–0.780)
	RD_P	0.688 (0.529–0.797)	0.689 (0.652–0.840)	0.689	0.719 (0.572–0.773)
	RD_EP	0.813 (0.653–0.899)	0.622 (0.529–0.692)	0.672	0.742 (0.606–0.804)
ROI-Based	RD_E	0.813 (0.708–0.901)	0.711 (0.602–0.746)	0.738	0.779 (0.733–0.847)
	RD_P	0.688 (0.583–0.842)	0.578 (0.445–0.710)	0.607	0.700 (0.574–0.785)
	RD_EP	0.938 (0.886–0.952)	0.733 (0.590–0.859)	0.787	0.892 (0.853–0.922)

The DeLong test p-values for all ROI-based combined models (R_E, R_P, R_EP, RD_E, RD_P, and RD_EP) are shown in Fig. 8. For the three models using only traditional radiomics features (R_E, R_P and R_EP), the AUC of the combined model (R_EP) is significantly higher than that of the individual early and peak phase models (p-values are 0.0079 and 1e-05, respectively). For the three combined models using both radiomics and deep learning features (RD_E, RD_P and RD_EP), the AUC of the combined model (RD_EP) is significantly higher than

that of the individual early and peak phase models (p-values are 1.5e-05 and 1.5e-11, respectively). The optimal model (RD_EP) has an AUC significantly higher than all other models (all p-values are less than 0.05).

Performance using the external validation dataset

The generalizability of the models was assessed using an external validation dataset. Since the ROI-based models consistently outperformed the image-based models, only the former was used for external validation. The results for each model on the external validation dataset are shown in Table 4.

Figure 9 shows the ROC curves for each model using the external validation dataset. The AUC values for R_E, R_P, R_EP, RD_E, RD_P, and RD_EP are 0.683, 0.698, 0.810, 0.772, 0.741 and 0.825, respectively. All models demonstrate good generalizability. The best-performing model remains the combined RD_EP model, with a Sensitivity of 0.778 (95% CI: 0.667–0.889), Specificity of 0.619 (95% CI: 0.526–0.706), Accuracy of 0.667 (20/30), and AUC of 0.825 (95% CI: 0.713–0.886).

Discussion

In this study, we extracted traditional radiomics features and 3D deep learning features from early and peak phase of DCE-MRI images. We integrated these features and selected the top 10 most discriminative features. Logistic regression model was used to predict pCR in breast cancer patients following NAT. The results show that when using ROI-based features, the model that integrates radiomics and deep learning features from early and peak phases (RD_EP) performs the best, with an accuracy of 0.787 and an AUC of 0.892.

Clinical significance of predicting pCR in BC-NAT patients

Breast cancer is one of the most common malignant tumors in women, and neoadjuvant therapy is a commonly used treatment method in clinical practice [1–3]. Pathological complete response is an effective and valuable surrogate endpoint for post-treatment survival. Accurately assessing the efficacy of neoadjuvant therapy is crucial in both clinical practice and fundamental research.

Patients who achieve pCR through NAT can obtain more favorable long-term prognoses [48]. However, some patients respond poorly to NAT (non-pCR), and these patients are unlikely to benefit from NAT. But these patients still suffer from the side effects of chemotherapy, targeted drugs, endocrine therapy, and radiotherapy. The adverse reactions of NAT may increase the physical burden on patients, leading to more complex subsequent treatment plans [49]. Furthermore, some patients may experience disease progression and tumor worsening during NAT treatment. Therefore, if the efficacy of

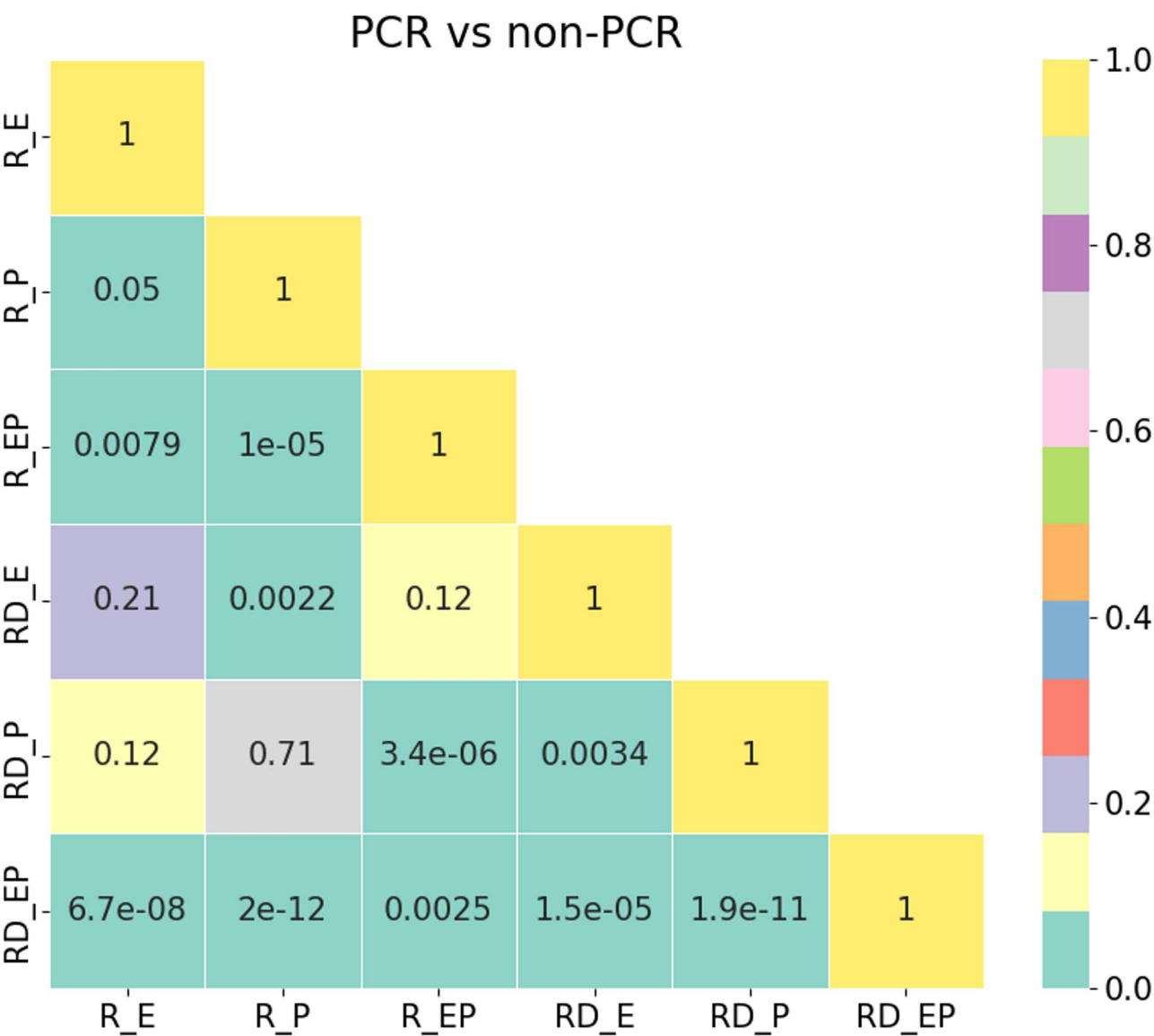


Fig. 8 DeLong test p-values for all ROI-based models

NAT in breast cancer patients could be non-invasively predicted before treatment, it would allow for maximally shrinking the tumor in patients likely to achieve pCR, thereby improving surgical success rates and long-term prognosis. For patients who are unlikely to achieve pCR, it would help avoid the side effects of ineffective chemoradiotherapy, allowing timely adjustments to the treatment strategy and improving the overall treatment outcomes for these patients.

It is often difficult for radiologists to determine whether the response to NAT is pCR or non-pCR from the patient’s pre-treatment or even post-treatment MRI images, which can affect treatment decisions and patient prognosis. The model developed in this study predicts the efficacy of NAT in breast cancer patients with an accuracy of up to 78.7%. The methods provided in this

study offer valuable support to clinicians, enabling them to quickly and accurately assess the benefit of NAT for patients. This allows for the development of personalized and precise treatment plans, minimizes unnecessary treatments, reduces the burden and side effects for patients, and ultimately improves treatment outcomes and long-term prognosis.

Feature selection and identified discriminative features

The initial number of extracted features in this study was indeed large, inevitably introducing a risk of multicollinearity. To address feature redundancy, we originally employed independent sample t-tests combined with LASSO. We further incorporated the minimum-redundancy maximum-relevance (mRMR) method, resulting in a revised pipeline (t-test → mRMR → LASSO). However,

Table 4 Results of the models on the external validation dataset

Combination	Model	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy	AUC (95% CI)
Single models	R_E	0.667 (0.545–0.800, 0.545, 0.800)	0.619 (0.533–0.714)	0.633	0.683 (0.566–0.790)
	R_P	0.778 (0.667–0.889)	0.571 (0.476–0.667)	0.633	0.698 (0.534–0.832)
	R_EP	0.889 (0.800–1.000, 0.800, 0.000)	0.619 (0.524–0.714)	0.700	0.810 (0.720–0.898)
Combined models	RD_E	0.889 (0.800–1.000, 0.800, 0.000)	0.571 (0.478–0.667)	0.667	0.772 (0.581–0.872)
	RD_P	0.667 (0.545–0.800, 0.545, 0.800)	0.667 (0.579–0.750)	0.667	0.741 (0.696–0.889)
	RD_EP	0.778 (0.667–0.889)	0.619 (0.526–0.706)	0.667	0.825 (0.713–0.886)

the additional step did not improve model performance and in some cases even led to a slight decline.

We believe this may be due to several reasons: (1) given the relatively small sample size, applying mRMR after the t-test—which had already removed a large number of irrelevant or noisy features—may have caused “over-filtering” and information loss; (2) since LASSO inherently manages multicollinearity, additional redundancy filtering may have unnecessarily restricted the candidate feature space and weakened LASSO’s ability to combine complementary features; and (3) as mRMR relies primarily on mutual information and correlation, it may excessively penalize nonlinear complementary information,

particularly in small-sample settings, thereby discarding potentially valuable features.

Conducting independent-sample t-tests may increase the risk of false positives. In our workflow, however, the t-test was used solely as a coarse, non-inferential screen to remove clearly irrelevant features. The final feature set and model were determined by LASSO with cross-validation, which explicitly penalizes model complexity and reduces redundancy. We therefore believe that this two-stage design substantially mitigates the risk of spurious findings, and that additional multiple-testing correction at the screening step would provide limited incremental value in the present setting.

Figure 10 shows the SHAP value analysis of features used in the six ROI-based models. A higher SHAP value for a feature indicates a greater contribution of that feature to the model’s output, while a lower SHAP value indicates a lesser contribution [47].

For traditional radiomics features, the GLSZM features from early-phase DCE-MRI and the GLRLM features from peak-phase DCE-MRI are of significant importance for predicting patient pCR, as evidenced by our SHAP analysis results. In the top-performing model RD_EP, the early-phase feature squareroot_glszm_GrayLevelNon-UniformityNormalized and the peak-phase feature wavelet-HLH_glrlm_GrayLevelNonUniformityNormalized have the largest contributions to the model, and they also contribute significantly to several other models. Most of the remaining features are GLSZM and GLRLM features, which also have notable contributions to the model predictions. Gray Level Size Zone Matrix (GLSZM) analyzes the distribution of gray level zones to reflect the texture features of a tumor, describing the texture uniformity and tissue structure in the tumor image. Gray Level Run Length Matrix (GLRLM) reflects the three-dimensional

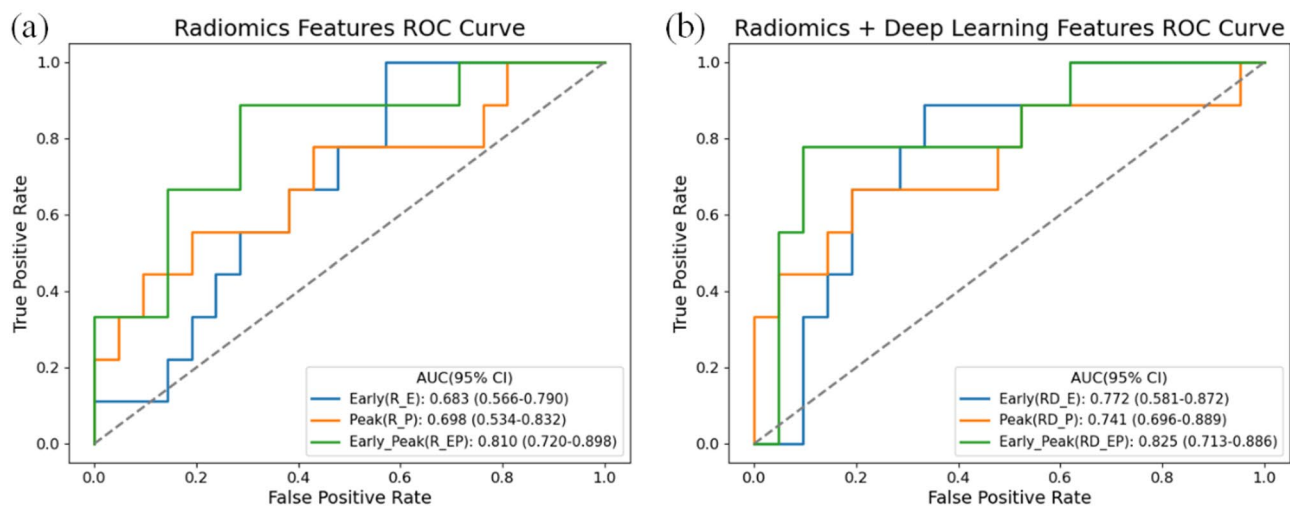


Fig. 9 ROC curves for ROI-based models using the external validation dataset. **a** Models using only radiomics features; **b** Combined models using radiomics and deep learning features

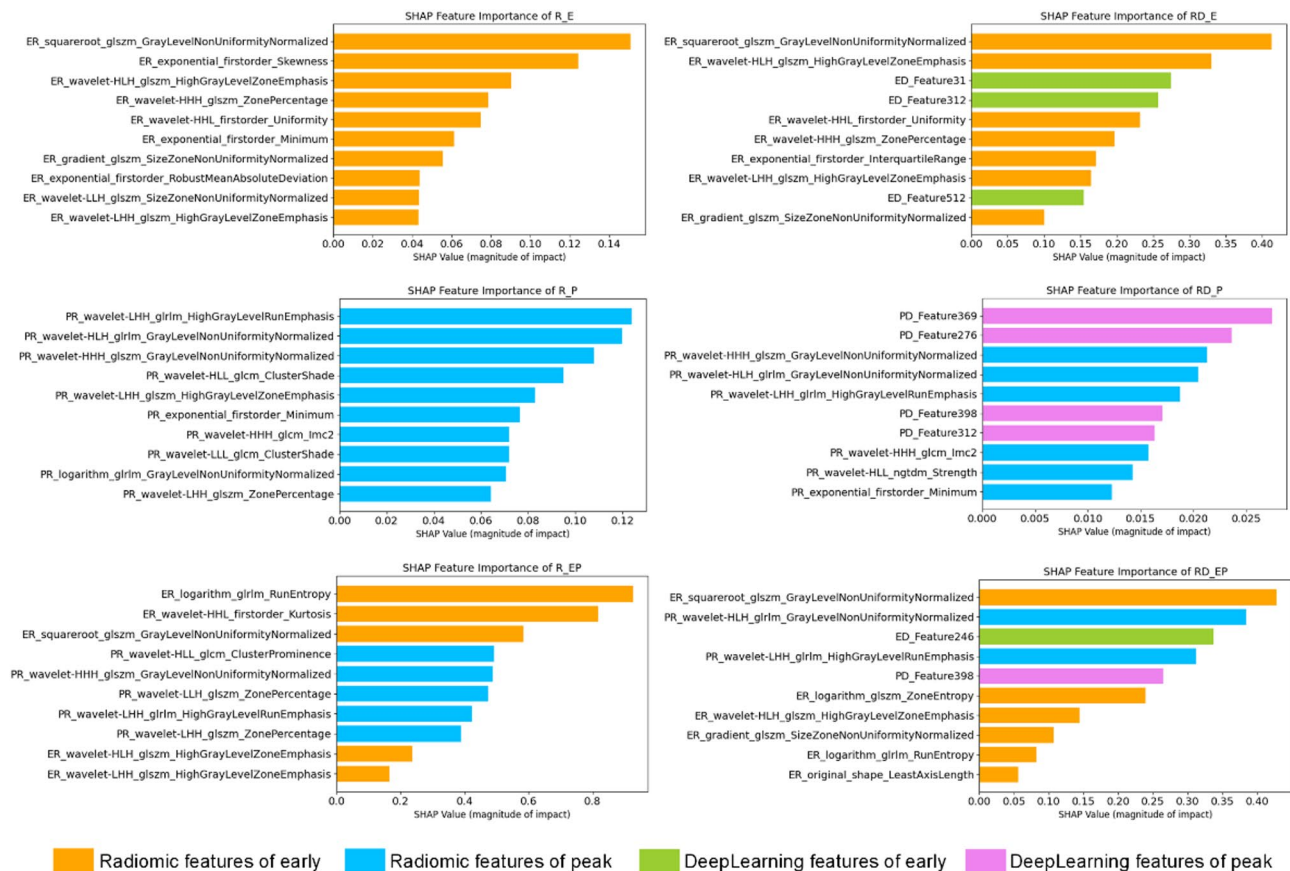


Fig. 10 Importance of different features in all models using SHAP analysis

structural characteristics of the tumor by analyzing the size and distribution of different gray level runs, describing the regional heterogeneity and structural complexity in the tumor image. The mean values of GLSZM and GLRLM are higher in the pCR group compared to the non-pCR group, which may indicate that tumors in patients achieving pCR exhibit greater internal structural complexity and heterogeneity [48, 50]. From Table 1, we observe that the pCR group in Dataset 1 has significantly lower ER and PR expression levels compared to the non-pCR group, while the proportion of Ki-67 is higher in the pCR group. This may suggest a close correlation between the high heterogeneity and complexity of tumors in pCR patients and the biological markers ER, PR, and Ki-67 expression levels. Previous studies have indicated that patients with HER2-positive and triple-negative breast cancer, who have low ER and PR expression, are more likely to achieve pCR after neoadjuvant therapy [51]. Studying the differences in these features among different patients may help to gain deeper insights into the biological characteristics and clinical behavior of tumors, thereby improving the precision and accuracy of radiological diagnosis.

For the deep learning features extracted using 3D-ResNet18, their high-dimensional and complex nature makes it challenging to interpret the detailed meaning of each feature. This difficulty primarily stems from the “black box” nature of deep learning models and the abstractness of the features themselves [52, 53]. However, from the SHAP analysis results, deep learning features from the early and peak phases of DCE-MRI also contribute to predicting pCR in breast cancer patients. These features may capture complex patterns and relationships in the imaging data that are difficult to discern. It is undeniable that deep learning features play a role in predicting pCR, a conclusion that has been confirmed in previous studies [35, 54].

Performance of models using combined features

We found that whether using traditional radiomics features alone or integrating radiomics and deep learning features, the model results that integrate early and peak phase features consistently outperform the results from early or peak phase alone. Moreover, whether for early phase, peak phase, or their combination, models that integrate traditional radiomics features with deep learning features show improved results compared to models

that use radiomics features alone. In this study, the model constructed using traditional radiomics and deep learning features from early and peak phases (RD_EP) yielded the best results, achieving an accuracy of 78.7% and an AUC of 0.892.

Additionally, we performed decision curve analysis (DCA) to evaluate the clinical utility of the radiomics-only models and the combined-feature models (Fig. 11). For the models using radiomics features alone, the R_EP model achieved a higher net benefit across a wider range of threshold probabilities than R_E and R_P, indicating superior support for clinical decision-making. In the

combined-feature models, RD_EP likewise outperformed RD_E and RD_P, providing higher net benefits over a broader threshold range. Moreover, RD_EP showed better clinical utility than R_EP in the radiomics-only group, suggesting that integrating multiple types of features can enhance a model's value in guiding clinical interventions.

Comparison of results between early and peak phases

In previous studies, early phase [22, 55–57] and peak phase [23, 56, 58] are the most commonly used for extracting radiomics features from DCE-MRI. In this study, we found that models using early phase images

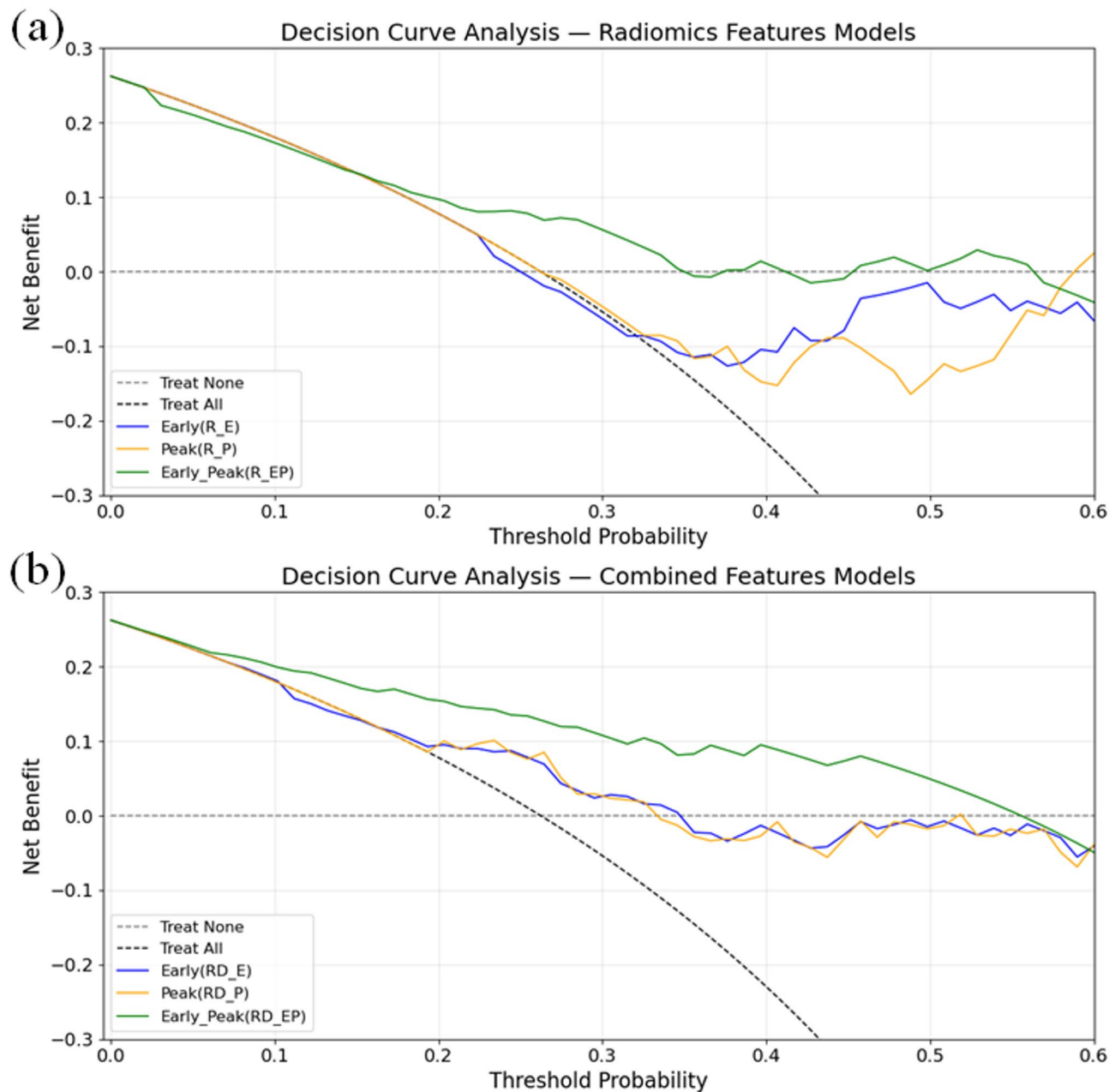


Fig. 11 Decision Curve Analysis (DCA) (a) Models using only radiomics features; (b) Combined models using radiomics and deep learning features

generally yielded better results compared to those using peak phase images. Previous studies have indicated that the early phase is the optimal stage for extracting reliable DCE-MRI dynamic parameters [59], and it is recommended for monitoring treatment response and predicting pCR after chemotherapy [16]. Our experimental results demonstrate that early-phase DCE-MRI outperforms peak-phase DCE-MRI in predicting pCR in breast cancer patients.

Breast cancer lesions usually exhibit enhancement primarily from the periphery towards the center. This pattern may be due to the higher density of tumor cells and microvessels in the peripheral regions compared to the central areas of the tumor, leading to more active biological behavior and greater invasiveness. In early DCE-MRI, the contrast agent rapidly enters the peripheral regions of the tumor lesion. At this stage, the extracted imaging features more accurately reflect the biological behavior of the tumor itself.

Furthermore, the early phase captures dynamic perfusion characteristics, including tumor vascular permeability, microvessel density, and initial blood flow patterns, which are closely associated with tumor aggressiveness and cellular activity. These parameters provide insights into the tumor microenvironment and its responsiveness to therapy. In contrast, the peak phase represents maximal contrast enhancement, which may be influenced by delayed wash-in and wash-out kinetics, edema, and necrotic regions, potentially diluting the specificity of the extracted features for predicting treatment response.

Therefore, the early-phase DCE-MRI images provides the most informative tumor characteristics, which are related to subsequent treatment outcomes and can aid in predicting pathological complete response (pCR) in breast cancer patients [60]. However, the value of peak-phase DCE-MRI images should not be overlooked either. During the clinical diagnosis process, the peak-phase DCE-MRI images often represent the optimal phase for lesion visualization, offering the highest contrast, which is associated with the complete penetration of contrast agent throughout the lesion and the achievement of peak blood perfusion. In this study, the image-based combined model RD_P demonstrated higher AUC, sensitivity, accuracy, and specificity than RD_E. Similarly, in the external validation set, the single-model R_P exhibited higher AUC and sensitivity than R_E. Thus, peak phase images of DCE-MRI contain equally rich information as early phase images, warranting further exploration and validation.

Methodological advantages

Table 5 presents a comparison of our method with previous studies. Our approach has the following three main advantages. Firstly, our study used both early and peak

Table 5 Comparison of our methods to those of previous studies

Study	Key aspects	Sensitivity	Specificity	Accuracy	AUC
Our Method	Early & Peak phase fusion; Radiomics & Deep Learning; T-test & LASSO.	0.938	0.733	0.787	0.892
Eun et al. [22]	Texture analysis; Random Forest feature selection.	0.625	0.917	0.831	0.820
Zheng et al. [24]	Three-region feature fusion; Elastic Net feature selection.	0.800	0.840	0.825	0.861
Sutton et al. [26]	Pre- & post-NAC texture features; mRMR & Elastic Net feature selection.	0.770	0.690	-	0.830
Peng et al. [35]	Image & Kinetics & molecular; Machine & deep learning.	0.781	0.769	0.772	0.832
Yu et al. [53]	Pre- & post-NAC feature fusion; Pearson & LASSO feature filter.	-	-	-	0.877

phase DCE-MRI images to predict pCR in breast cancer patients, demonstrating superior predictive accuracy and potential. Liu et al. [23] only used images from a single phase for prediction. Compared to studies using images from a single phase (typically early or peak), to analyze both early and peak-phase images allows for a more comprehensive capture of tumor dynamics, providing additional information about the tumor and thereby enhancing predictive accuracy.

Secondly, in our study, we not only developed traditional radiomics models but also constructed combined models integrating both radiomics and deep learning features. This approach significantly improved the accuracy of predicting pCR in breast cancer patients. We demonstrated that the combined model incorporating deep learning features outperforms traditional radiomics models. Peng et al. [35] also applied deep learning methods to predict the pathological response of breast cancer to NAC and found that these methods performed better than conventional radiomics approaches. Deep learning algorithms can automatically extract complex, high-dimensional features, capturing tumor information that traditional radiomics features might overlook. By integrating traditional radiomics and deep learning features, we can leverage the strengths of both methods to enhance the accuracy of pCR prediction in breast cancer patients.

Finally, the deep learning network employed in our study was 3D-ResNet18, a three-dimensional convolutional neural network capable of capturing subtle variations and complex structures in MRI images. Li et al. [61] utilized a 2D UTransNet primarily for deep learning feature extraction from two-dimensional MRI images, which may lack comprehensive capture of the three-dimensional tumor information present in MRI scans.

Limitations and future work

This study has several limitations. First, both the dataset used for model development (Dataset 1) and the dataset used for external validation (Dataset 2) have relatively small sample sizes. We aim to collaborate with more hospitals and use larger sample sets in future research to validate the stability and generalizability of our model. Second, our study relied solely on MRI data for predictions. Although MRI provides excellent anatomical and functional details, other imaging modalities can offer complementary information. For instance, ¹⁸F-FDG PET/CT captures the metabolic phenotype of tumors, and recent studies have demonstrated that AI models based on PET/CT radiomics can achieve strong performance in predicting pCR [62]. Furthermore, the efficacy of neoadjuvant therapy in breast cancer patients is influenced by a combination of imaging features, pathological characteristics, and clinical factors. Notably, future research

should focus on developing integrated predictive models that incorporate multimodal imaging data (e.g., MRI and PET/CT) together with pathological and clinical features. Third, in feature fusion, we only performed a simple concatenation of features followed by feature selection. Future studies could explore more advanced feature fusion techniques and model ensemble strategies to further optimize predictive performance and enhance model stability. Finally, although the optimal model in our study achieved an accuracy of 78.7%, over 20% of breast cancer patients were still predicted incorrectly. This could have significant implications for these patients. Future work should aim to further refine the model and improve its predictive capabilities, potentially through the application of advanced deep learning networks.

Conclusions

In summary, this study utilized traditional radiomics features and 3D deep learning features from early and peak phase of DCE-MRI, employing logistic regression algorithms to predict pCR in breast cancer patients. Our results demonstrate that integrating different phases of DCE-MRI enhances the accuracy of predicting pCR in breast cancer patients. Additionally, the approach of integrating traditional radiomics features with 3D deep learning features shows considerable potential in predicting pCR. This offers new perspectives for future research and clinical applications, particularly in personalized treatment and prognostic assessment. By combining features extracted from multi-phase DCE-MRI images, clinicians can more accurately predict patient responses to treatment, allowing for the development of more effective treatment strategies and improved patient outcomes.

Abbreviations

AUC	Area Under the Curve
CI	Confidence Interval
DCE-MRI	Dynamic contrast-enhanced magnetic resonance imaging
ER	Estrogen receptor
GLCM	Gray level co-occurrence matrix
GLDM	Gray level dependence matrix
GLSZM	Gray level size zone matrix
GLRLM	Gray level run length matrix
HER-2	Human epidermal growth factor receptor 2
LASSO	Least absolute shrinkage and selection operator
LR	Logistic regression
MRI	Magnetic resonance imaging
MSE	Mean squared error
NAT	Neoadjuvant therapy
NGTDM	Neighboring gray tone difference matrix
non-pCR	Non-pathological complete response
pCR	Pathological complete response
PET-CT	Positron emission tomography – computer tomography
PR	Progesterone receptor
ResNet	Residual neural network
ROC	Receiver operating characteristic
ROI	Region of interest
R_E	Radiomics Features - Early phase Model
R_P	Radiomics Features - Peak phase Model
R_EP	Radiomics Features - Early and Peak phases Model
RD_E	Radiomics and Deep Learning Features - Early phase Model

RD_P	Radiomics and Deep Learning Features - Peak phase Model
RD_EP	Radiomics and Deep Learning Features - Early and Peak phases Model.
SHAP	Shapley additive explanations
SMOTE	Synthetic minority over-sampling technique
TNM	Tumor-node-metastasis

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Not applicable.

Authors' contributions

YZ performed experiments and analyzed the data along with JC. SQ and DZ conceived the study, presented the results, and wrote the manuscript along with YZ. JC, CC and DZ collected and analyzed the data. All authors read and approved the final manuscript.

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Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The Medical Ethics Committee of Liaoning Cancer Hospital and The Affiliated Hospital of Qingdao University approved this retrospective study (No. 20221101; QYFY WZLL 28631), and the informed consent was waived. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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